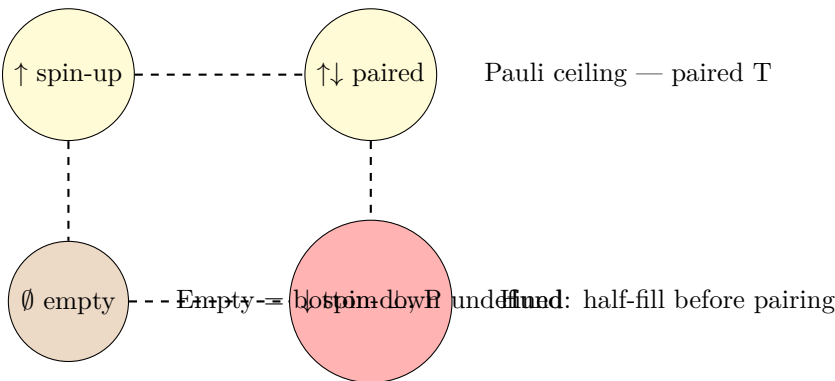


Standard Model Hierarchy: Topological State-Filling Model

Full Extension from Original Diagrams

June 2026

1 Z2 Spin Structure



2 Z3 Quarks and SU(3) Gluons

The quark color layer adds 3-antichain $\{R, G, B\}$ with forced collapse to singlet. Gluon tree: λ_0 (excluded) $\rightarrow \lambda_3, \lambda_8$ (diagonal) $\rightarrow 6$ off-diagonal $\rightarrow$ Cartan $\rightarrow$ vacuum.

Gell-Mann basis: $\lambda_3 = (R\bar{R} - G\bar{G})/\sqrt{2}$ $\lambda_8 = (R\bar{R} + G\bar{G} - 2B\bar{B})/\sqrt{6}$ (and full set for λ_{1-7} as in original images).

3 Full Standard Model

Layer	Group	Key Features
Z2	Spin/Weak	Pauli ceiling, Hund rule, square graph
Z3	Color	Antichain confinement to singlet
SU(3) adj	Gluons	8 bosons, Cartan + off-diagonal, non-Abelian
SU(2)xU(1)	EW	3W + B, Higgs breaking to photon + massive
Fermions	3 gen	45 Weyl fields, Yukawa pairing
Higgs	Scalar doublet	Eats Goldstones, mass generation

Probability flows and topological antichains unify the entire structure as described in the report text above.